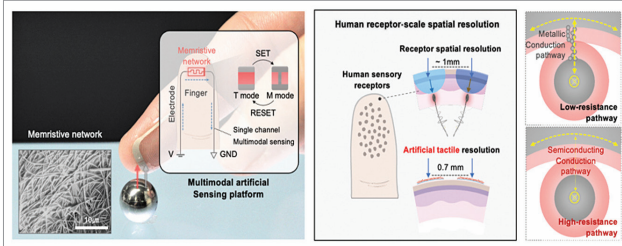


Artificial skin enables human-like robotic sensing

Researchers at Seoul National University have developed an ultrathin artificial skin that enables robots to sense both temperature and pressure through a single flexible sensor,

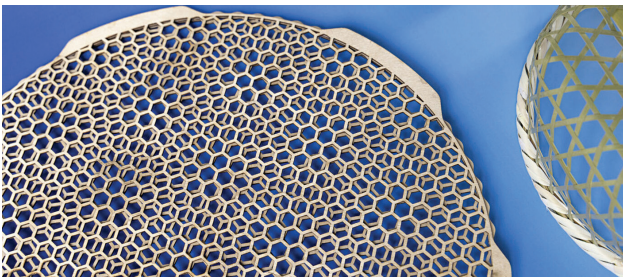


Operating principles and key characteristics of the multimodal artificial sensing platform (Credit: Nature Materials)

advancing tactile perception for Physical AI systems. Developed in Seoul, the multimodal sensor uses a silver-core, copper oxide-shell nanowire architecture that switches between thermal and mechanical sensing 16 times per second with rapid response times. The technology achieved up to 95 % classification accuracy by combining both sensing modes and identified everyday objects with 83 % accuracy. It could enhance prosthetics, wearable electronic skin, soft robotics, robotic grippers, and human-machine interfaces.

Machine learning helps discover two new superconductors

A team at Aalto University has discovered two new superconductors, YRu_3B_2 and LuRu_3B_2 , using a machine-learning approach combined with quantum physics calculations to accelerate materials discovery. Developed in Finland under the SuperC consortium, the materials exhibit superconductivity



YRu_3B_2 and LuRu_3B_2 gain their superconductivity from atoms arranged in a kagome lattice, named after a hexagonal Japanese basket-weaving pattern (Credit: <https://www.aalto.fi/en>)

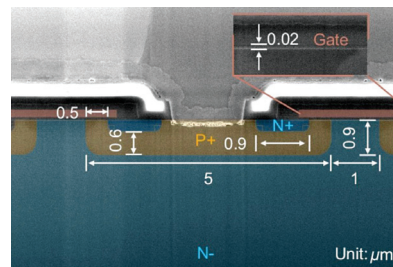
through a kagome lattice electron structure. Machine learning identified promising candidates, which were then validated through theoretical modelling and experiments. The approach could dramatically speed the search for room-temperature superconductors, enabling future advances in quantum computing, MRI systems, fusion reactors, magnetic levitation transport, and energy-efficient computing and data centres.

Quantum device helps prevent GPS spoofing

The Commonwealth Scientific and Industrial Research Organisation has developed two portable quantum light sources that bring quantum-secured timing closer to real-world deployment, helping maintain accurate timing when GPS signals are jammed, spoofed, or unavailable. Engineered by Australia's national science agency as part of a Defence Science and Technology Group-led project, the devices generate entangled photon pairs to detect interference during ground-to-satellite time transfer. Any attempt to intercept the signal alters its quantum state, exposing tampering instantly. The technology could strengthen secure timing for defence, telecommunications, power grids, transport, banking, and emergency services.

'Brain-like' chip to advance quantum computing

The University of Hong Kong (HKU) has developed a programmable neuromorphic hardware platform that operates



HKU Engineering develops 'Brain-Like' chip to advance quantum computing and deep-space exploration (Credit: <https://thequantuminsider.com>)

at temperatures as low as 10 millikelvin, enabling low-power control electronics for quantum computers. Developed in Hong Kong, the platform uses standard silicon carbide (SiC) MOSFETs that mimic neuron-like spiking behaviour through a stable negative differential resistance mechanism. The technology can form cryogenic neural networks for local data processing, improving quantum error correction while reducing heat and wiring complexity. It could also support deep-space electronics and be manufactured using existing 300mm semiconductor fabrication processes.

Graphite found useful for quantum and superconducting tech

A team at Massachusetts Institute of Technology (MIT) has discovered that naturally occurring rhombohedral graphene within graphite can support four distinct superconducting states, offering new insights into unconventional superconductivity. Conducted in the United States, the study showed that three states remained stable under magnetic fields of up to 9 tesla, while another became stronger, with its transition temperature rising from about 55mK to nearly 90mK. The findings suggest unconventional electron pairing mechanisms and establish naturally occurring graphite as a promising platform for quantum materials research and future superconducting technologies.